NOTE The document identifier and heading have been changed on this page to reflect that this is a performance specification There are no other changes to this document The document identifier on subsequent pages has not been changed, but will be changed the next time this document is revised

INCH-POUND

MIL-PRF-83532/5A
30 December 1991
SUPERSEDING
MIL-D-83532/5
10 March 1987

PERFORMANCE SPECIFICATION SHEET

DELAY LINES, ACTIVE,
16-PIN DIP COMPATIBLE,
PROGRAMMABLE 3-BIT, TTL COMPATIBLE

This specification is approved for use by all Departments and Agencies of the Department of Defense

The requirements for acquiring the product described herein shall consist of this specification sheet and the issue of the following specification listed in that issue of the Department of Defense Index of Specifications and Standards (DODISS) specified in the solicitation MIL-D-83532

REQUIREMENTS

Design and construction Parts shall be of the design, construction, and physical dimensions specified in the appendix of MIL-D-83532 Two case styles (F and F) are available

Input pulse Delay lines must be capable of meeting applicable table I and table II requirements with an input pulse having the following characteristics

A Leading edge of a positive-going pulse

B Minimum pulse width of 50 percent of maximum delay time

C Fixed pulse repetition rate equal to ten times the maximum delay time

D Duty cycle not to exceed 50 percent

Schematic See figure 1

Delay times Delay time from input to all taps shall be as specified in table I (+25°C) and table II (-55°C and +125°C)

Output rise time (applied to leading edge and trailing edge) 4 ns maximum Measurement conditions (-55°C ≤ T_C ≤ +125°C) V_{CC} = 5.0 V dc TR_I ≤ 3 ns

DC parameters (over operating temperature range) See table III

Truth table See table IV

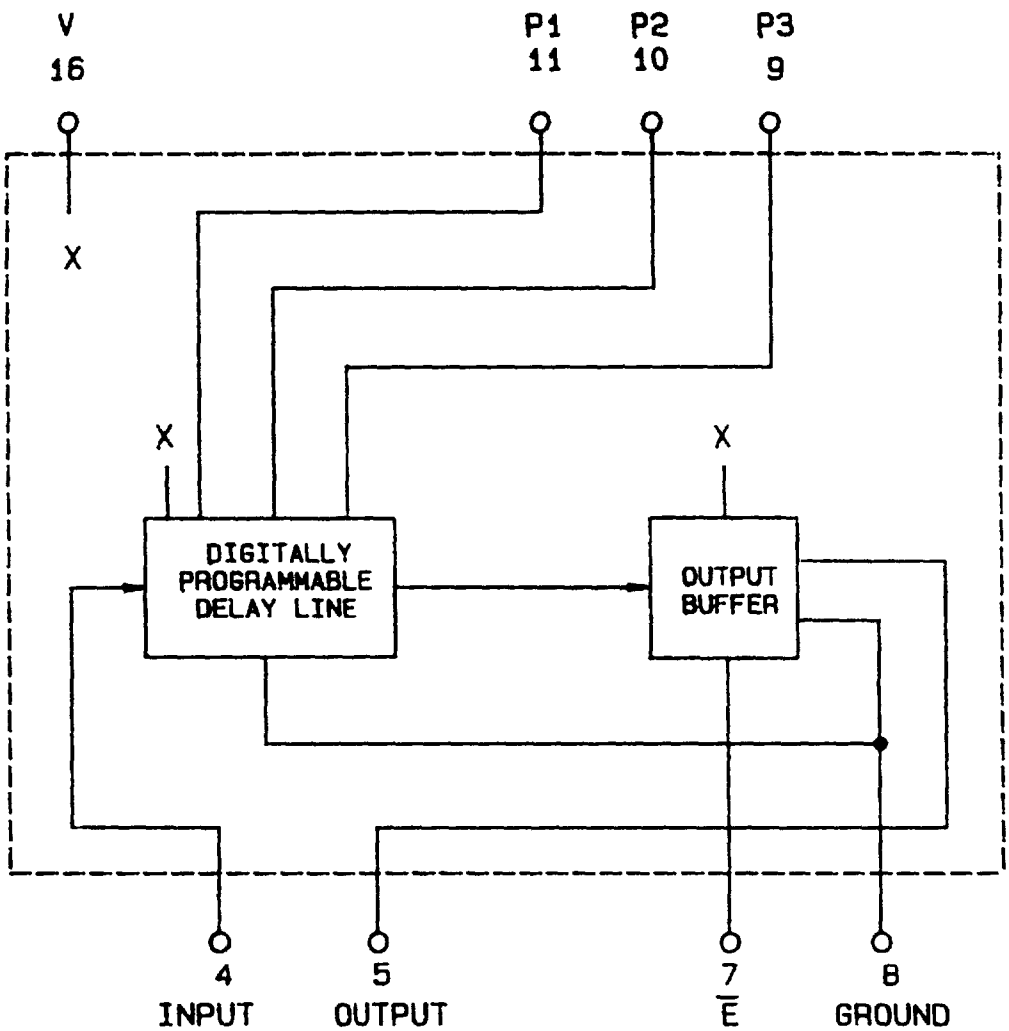
Rated maximum load (fan-out) Ten TTL Schottky loads per tap (no more than 20 TTL Schottky loads per unit)

Terminations Part terminations shall be printed-circuit pin type in accordance with MIL-H-38534

Operating temperature Operating temperature range shall be -55°C to +125°

DISTRIBUTION STATEMENT A Approved for public release; distribution is unlimited

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NOTES
1. Schematic is for general information only
2. Pins 1, 2, 3, 6, 12, 13, 14, and 15 are omitted on level A delay lines
3. All pins are installed on level B delay lines

FIGURE 1 Schematic

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TABLE I Dash numbers and delay characteristics at +25°C, V_{CC} = 5.00 ± .01 volts

1/ Dash number	2/ 3/ Step zero delay time (ns)	3/ Maximum delay time (ns) (+2.0 ns or 5% whichever is greater)	3/ Delay change per step (ns)	4/ Maximum deviation from programmed delay (ns)
001	6.0 ±1.5	9.5	0.5 ±0.3	±1.0 or 5% (whichever is greater)
002	"	13	1 ±0.4	"
003	"	20	2 ±0.8	"
004	"	27	3 ±1.0	"
005	"	41	5 ±1.5	"
006	"	76	10 ±1.5	"
007	"	111	15 ±1.5	"
008	"	146	20 ±1.5	"
009	"	286	40 ±2.5	"
010	"	356	50 ±3.5	"

1/ For PIN, maximum delay time, tabulation and truth table examples, a step zero delay time of 6 ns was assumed

2/ Delay at step zero is referenced to the input pin

3/ The delay is being measured at +1.5 volt level, with no load

4/ All delay times after step zero are referenced to step zero

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TABLE II. Dash numbers and delay characteristics at -55°C and +125°C, V_{CC} = 5.00 ± .01 volts

1/ Dash number	2/ 3/ Step zero delay time (ns)	3/ Maximum delay time (ns) (±4.0 ns or 8% whichever is greater)	3/ Delay change per step (ns)	4/ Maximum deviation from programmed delay (ns)
001	6.0 ±3.5	9.5	0.5 ±0.6	±2.0 or 8% (whichever is greater)
002	"	13	1 ±0.8	"
003	"	20	2 ±1.6	"
004	"	27	3 ±2.0	"
005	"	41	5 ±2.5	"
006	"	76	10 ±3.0	"
007	"	111	15 ±3.0	"
008	"	146	20 ±3.5	"
009	"	286	40 ±4.5	"
010	"	356	50 ±6.0	"

1/ For PIN, maximum delay time, tabulation and truth table examples, a step zero delay time of 6 ns was assumed

2/ Delay at step zero is referenced to the input pin

3/ The delay is being measured at +1.5 volt level, with no load

4/ All delay times after step zero are referenced to step zero

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TABLE III DC characteristics.

Test	Symbol	Conditions $-55°C \leq T_C \leq +125°C$	Limits		Unit
			Min	Max	
High level output voltage	V_{OH}	V_{CC} = 4 75 V V_{IH} = 2 0 V I_{OH} = -1 mA	2 5		V
Low level output voltage	V_{OL}	V_{CC} = 4 75 V V_{IL} = 0 8 V I_{OL} = 20 mA		0 5	V
Input clamp voltage	V_{IC}	V_{CC} = 4 75 V I_I = -18 mA T_C = +25°C		-1 2	V
High level input current	I_{IH1}	V_{CC} = 5 25 V, V_{IH} = 2 5 V		2 0	mA
	I_{IH2}	V_{CC} = 5 25 V, V_{IH} = 5 25 V		43	mA
Low level input current	I_{IL}	V_{CC} = 5 25 V, V_{IL} = 0.5		-36 0	mA
Short circuit output current	I_{OS}	V_{CC} = 5 25 V, V_{OS} = 0 0 V (not more than one output shorted at a time)	-40	-100	mA
Low level supply current	I_{CCL}	V_{CC} = 5 25 V V_I = 0 0 V		100	mA

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TABLE IV Truth table

Programming pins	3 2 1	0 0 0	0 0 1	0 1 0	0 1 1	1 0 0	1 0 1	1 1 0	1 1 1
Dash numbers									
001		6	0 5	1	1 5	2	2 5	3 0	3 5
002		6	1	2	3	4	5	6	7
003		6	2	4	6	8	10	12	14
004		6	3	6	9	12	15	18	21
005		6	5	10	15	20	25	30	35
006		6	10	20	30	40	50	60	70
007		6	15	30	45	60	75	90	105
008		6	20	40	60	80	100	120	140
009		6	40	80	120	160	200	240	280
010		6	50	100	150	200	250	300	350

Part or Identifying Number (PIN) The PIN shall be in the following format

M	83532/05	A	001	B	C

Military designator

Specification sheet number

Case style E or F (see MIL-D-83532 appendix

Dash number (see table I)

Product assurance level A, B, or S (see MIL-D-83532)

Lead finish (see MIL-D-83532)

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QUALITY ASSURANCE PROVISIONS

Extent of qualification. The extent of qualification shall be as specified in MIL-D-83532, for example

Qualification and testing of M83532/05A001A* 1/ and M83532/05A010A* 1/ shall be sufficient to grant qualification to all dash numbers with case style E contained in this specification sheet Similarly, for case style F, qualification and testing of dash numbers 001 and 010 shall be sufficient to grant qualification to all dash numbers in this specification sheet having the same case style as the components tested

Production assurance levels Product assurance level A shall not extend to level B Level B shall not extend to level A

Supersession data· MIL-D-83532/5 supersedes DESC drawing number 85019 when a qualified source is available

Revision letters are not used to denote changes due to the extensiveness of the changes

1/ * - Asterisk represents any of the applicable finishes

CONCLUDING MATERIAL

Custodians
Army - ER
Navy - EC
Air Force - 85

Review activities
Air Force - 11, 19, 99
DLA - ES

User activities
Navy - AS, CG, MC, SH

Preparing activity
Air Force - 85

Agent
DLA - ES

(Project 5999-0284-05)

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